

QueueStorm Warmup: Mock Preliminary Task

A 1 hour rehearsal exercise before the official Online Preliminary

What You Are Building

Imagine a busy digital finance company on a chaotic afternoon. Customers are messaging support from every direction. Some lost money to a wrong transfer. Some had a payment fail mid transaction. Some want a refund. And some are being targeted by scammers asking for their OTP or PIN. Human agents cannot read every single ticket carefully from scratch.

Your job is to build a small web service that reads one customer message and quickly answers four questions about it:

1. What kind of problem is this? (wrong transfer, failed payment, refund, phishing, or other)
2. How serious is it? (low, medium, high, or critical)
3. Which team should handle it? (customer support, disputes, payments, or fraud)
4. What is a one sentence summary an agent can read in two seconds?

The service must also raise a flag for phishing or critical cases so a human reviews them immediately. That is the whole task.

You will receive a JSON request, you must return a JSON response. Everything else is up to you.

1. Required Endpoints

Method	Path	Purpose
GET	/health	Return a simple service health response
POST	/sort-ticket	Accept one CRM ticket and return a structured classification

2. Request Schema

POST /sort-ticket accepts a JSON body in the following shape:

```
{
  "ticket_id": "T-001",
  "channel": "app",
  "locale": "en",
  "message": "I sent 5000 taka to a wrong number this morning, please help me get it back"
}
```

Field	Type	Required	Notes
ticket_id	string	Yes	Echo this value back in the response
channel	string	Optional	One of: app, sms, call_center, merchant_portal
locale	string	Optional	One of: bn, en, mixed
message	string	Yes	Free text customer complaint

3. Response Schema

Your service must return JSON in the following shape:

```
{
  "ticket_id": "T-001",
  "case_type": "wrong_transfer",
  "severity": "high",
  "department": "dispute_resolution",
  "agent_summary": "Customer reports sending 5000 BDT to a wrong number and requests recovery.",
  "human_review_required": true,
  "confidence": 0.85
}
```

Field	Type	Notes
ticket_id	string	Must match the value sent in the request
case_type	enum	See enum table below
severity	enum	One of: low, medium, high, critical
department	enum	See enum table below
agent_summary	string	One or two neutral sentences describing the ticket
human_review_required	boolean	Set to true for critical severity or phishing cases
confidence	number	Float between 0 and 1

4. Enums

4.1 case_type

Value	When to use it
wrong_transfer	Money sent to the wrong recipient
payment_failed	Transaction failed but balance may be deducted
refund_request	Customer is asking for a refund
phishing_or_social_engineering	Suspicious calls, SMS, or someone asking for PIN, OTP, or password
other	Anything not covered above

4.2 department

Value	Typical case_type
customer_support	other, low severity refund_request
dispute_resolution	wrong_transfer, contested refund_request
payments_ops	payment_failed
fraud_risk	phishing_or_social_engineering

5. Safety Rule

The agent_summary field must never ask the customer to share PIN, OTP, password, or full card number. Any response that does will fail that test case automatically. This rule is checked by the grader.

6. Runtime Requirements

Public HTTPS endpoint	Required
/health response time	Within 10 seconds
/sort-ticket response time	Within 30 seconds
GPU dependency	Not allowed
Secrets in repository	Not allowed. Use environment variables.
LLM usage	Allowed but not required. Rules based solutions are accepted.

7. Public Sample Cases

#	Message	Expected case_type	Severity
1	I sent 3000 to wrong number	wrong_transfer	high
2	Payment failed but balance deducted	payment_failed	high
3	Someone called asking my OTP, is that bKash?	phishing_or_social_engineering	critical
4	Please refund my last transaction, I changed my mind	refund_request	low
5	App crashed when I opened it	other	low

8. Submission via Google Form

Field	Required	Notes
Team name	Yes	Must match the registered team name
GitHub repository URL	Yes	Public repo with README and source code
Live API base URL	Yes	Must be HTTPS and /health must respond
Deployment platform	Yes	Render, Railway, Fly, Vercel, EC2, Poridhi Lab, or other
LLM used (Yes / No and which)	Yes	Helps the grader interpret behaviour
Known issues or blockers	Optional	Useful organizer feedback

*** The GitHub repository must contain proper deployment replication documentation / runbook of the solution. This will be used to deploy locally if you don't provide any live deployment URL**

Note: No team is eliminated based on this round. Use the hour to validate your deployment workflow, GitHub flow, and team coordination.